Krish Sharma

Aspiring Software Engineer passionate about systems programming, distributed architectures, system design, and full-stack web applications.

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Education

Manipal Institute of Technology — Bengaluru, India
Bachelor of Technology in Computer Science Engineering – ***CGPA: 8.21/10.0*** — *Aug. 2023 – May 2027*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Compiler Design

Birla Public School Pilani — Pilani, India
Senior Secondary Education (Class XII) – ***Percentage: 91.8%*** — *2022*

Birla Public School Pilani — Pilani, India
Secondary Education (Class X) – ***Percentage: 90.8%*** — *2020*

Technical Skills

Languages: C++, Python, Java, C, JavaScript, TypeScript, SQL, HTML/CSS

Web & Backend: React.js, Node.js, Express, Flask, Socket.io, Yjs (CRDTs), REST APIs, Session Caching

Systems & Cloud: Redis, Multithreading, Thread-Safe Connection Pooling, Docker, Linux/Unix, MySQL, SQLite

AI / Data Science: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, NLP, LLM Integration (Groq API)

Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA, GDB, Postman, Streamlit Cloud

Projects

CodeCraft Studio – Collaborative IDE — *React, TypeScript, Monaco, Yjs, Socket.io, SQLite* **GitHub** | **Live Demo**

- Architected a zero-conflict real-time collaborative IDE leveraging Yjs CRDTs over WebSockets, guaranteeing eventual consistency and achieving **<35ms** state synchronization latency across 50+ concurrent client sessions.
- Integrated Monaco Editor with dynamic remote cursor tracking, active line indicators, and granular operational transformations, resulting in a **99.8%** conflict resolution efficiency during simultaneous edits.
- Engineered a secure code execution sandbox and version history engine backed by SQLite and RESTful services, processing multi-language execution requests with **<120ms** execution overhead.

Distributed Job Orchestrator — *C++, Redis, Multithreading, Connection Pooling, Docker* **GitHub** | **Live Demo**

- Built a high-throughput, fault-tolerant background task queue in C++20 using POSIX threads and thread-safe data structures, scaling job processing to **10,000+ jobs/sec** with zero deadlocks.
- Implemented an asynchronous pub/sub worker distribution model with Redis Streams and connection pooling, cutting connection acquisition latency by **65%** and achieving **99.9%** task execution reliability.
- Designed exponential backoff auto-retry policies and worker heartbeating, enabling automated failover recovery in **<500ms** and containerizing the daemon with Docker for isolated deployment.

Nexus Compiler – 4-Stage Language Compiler — *Python, Flask, Vanilla JS, x86 Assembly* **GitHub** | **Live Demo**

- Designed and implemented a full 4-stage compiler for a Pascal-like DSL, featuring a custom lexer, recursive descent parser, Three-Address Code (TAC) optimizer, and pseudo-x86 code generator.
- Implemented dead-code elimination and constant folding passes in intermediate representation (TAC), reducing generated x86 instruction count by **28%** and improving program execution speed by **18%**.
- Developed a real-time web visualization dashboard using Flask & Vanilla JS to render interactive Abstract Syntax Trees (AST) and symbol table state transitions with **<50ms** rendering latency.

Certifications

Ordered Data Structures – University of Illinois Urbana-Champaign (Coursera)

Introduction to Java and Object-Oriented Programming – University of Pennsylvania (Coursera)

ChatGPT Prompt Engineering for Developers – DeepLearning.AI

Python 101 for Data Science – IBM / Coursera Project Network